TOPOLOGY / ALGORITHMS / VERIFIED COMPUTATION

Exceptional Rows and Fast Certificate Replay

for Unknot Recognition

Exact forest selection, a Boolean determinant sieve, and authenticated interval contraction

Research continuation for the ProveIt project
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Abstract

Recent ProveIt work reduces one source-authenticated handle simplification problem to choosing a rooted dual spanning tree whose deleted rows leave a completely peelable unsigned occurrence matrix. Its polynomial theorem handles useful columns supported on at most two rows; native three-row columns remain outside that theorem. We close this parameterized gap.

Let U be the set of rows incident to the exceptional columns and put $t = |U|$. Ordinary propagation can be summarized at U. A fixed family of disjoint source-component requirements and a Boolean closure function on U completely characterize successful trees. A Möbius-weighted determinant sum retains exactly their copied common-basis monomials. Collapsing ordinary strongly connected components and restricting to the J ideals of their reachability poset gives one-sided randomized witness discovery in $J\,\mathrm{poly}(N)$ time, where $J \le 2^t \le 8^k$ for k native triple columns. If that poset has width w, a deterministic algorithm takes $J(m+1)^w\,\mathrm{poly}(N)$ time and polynomial space. Every positive answer has a deterministic independent tree-and-peeling check. An explicit homogeneous-polynomial lemma extracts a witness from one nonzero evaluation without further randomness. We also give a deterministic exceptional-schedule fallback and a Matrix–Tree counting alternative. An optimal lower forest minimizes the raw triple count by a graphic cycle-nullity formula.

The second contribution addresses measured certificate costs. An exact ordered-gap quotient and its inverse replace repeated prefix rescans by prefix tables in three independent verifiers. Endpoint transport changes from $O(pg)$ to $O(g + p\log(g+2))$ arithmetic work. Invocation-local reuse removes duplicate source reconstruction while preserving the distinction between producer data and independent verification. The package includes source patches, full-certificate comparison, native and finite-model audits, paired measurements, and executable research code. On the pinned 106-case corpus, paired full-query time falls by 23.5%; a separate bounded lower-forest portfolio supplies five additional whole-cut certificates.

These are local algorithmic and implementation results. They do not prove a general quasi-polynomial unknot-recognition bound. The article specifies the remaining geometric production, transport, and global hierarchy obligations and proposes concrete follow-up problems.

Reproducibility anchor. Repository baseline:
`c6d0ec8aa0b6308310ab6c081ba29d8bc363ede5`

Claim discipline. Finite-model theorems are proved. Empirical results concern the supplied corpus and recorded environment. Unfinished geometric and global-complexity steps are identified explicitly.

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1 What this continuation adds

The practical question is how to reduce the cost of verified topology operations while removing genuine combinatorial search barriers. The repository contains many useful exact operators; a new report must identify which obstacle it actually removes. This continuation contributes a new parameterized theorem for choosing dual forests and an independently tested improvement to interval-certificate replay. The theorem and implementation operate at different interfaces, which makes their scope straightforward to audit.

1.1 The inspected checkpoint

We inspected both requested directories at the pinned revision, including all five ZIP archives then present in `docs/incoming`. These contain component-gauge cut atlases, dual-forest normalization, dual forests with homological progress, flag regularization, and positive cancellation. They are listed with hashes in the package provenance file. Their results are antecedents, not new results of this continuation. In particular the following ingredients were already available:

1. exact column-leaf peeling and its greatest stopping-set obstruction;
2. source-authenticated removal of native three-handles by a rooted dual forest;
3. an exact tree/stopping-set criterion, and a polynomial graphic–partition matroid-intersection algorithm when each useful column has at most two incident rows;
4. the fact that an original native cocore has at most three actual occurrences, isolating $(1, 1, 1)$ as the remaining native support type;
5. saturated interval-bundle compilation and independently replayable whole-cut handlebody certificates.

The maintained whole-cut corpus has 106 cases. At this pin its existing producer returns 92 complete handlebody-census certificates and 14 inconclusive results. The behavior-preserving performance patch preserves that classification. The optional lower-forest portfolio adds five independently verified whole-cut positives. The separate forest research operator decides a more general supplied graph/matrix problem; it is not inserted into the native recognition schedule without a source adapter and a justified dispatch rule. The detailed measurements and fresh source checks appear in section 8.

1.2 A new parameter, followed by a stronger algorithm

The preceding two-row theorem is polynomial even when the dual graph has many vertices. Its limitation is the presence of columns that encode a genuine conjunction: two other rows must disappear before the third can peel. Treating either premise as sufficient would be unsound.

Our first extension guesses the exceptional activations and reduces their prerequisites to a bounded number of overlapping edge-cover requirements. This gives an exact deterministic algorithm, with the exceptional-column count in the exponent instead of the dual rank. It also gives a useful reference implementation and a transparent completeness proof.

A stronger reduction removes the need to guess an activation order. Only the rows occurring in nonlinear rules need to remember the influence of the tree. Everything else is ordinary directed reachability. The effect of a spanning tree on these exceptional rows is a finite bit mask, and successful masks form a monotone Boolean function that can be evaluated by forward chaining. We compile

that function into a determinant filter. Unlike an inclusion–exclusion sum that asks only whether every color appears, the filter accepts precisely the masks that permit complete nonlinear closure. Each valid copied common-basis monomial retains its original nonzero determinant coefficient: the filter multiplies it by one. Invalid monomials are multiplied by zero.

Result	Time, ignoring polynomial factors	Exact scope
Preceding two-row theorem	1	Every useful column has support at most two.
Deterministic extension	$3^k k!(m+1)^{2k}$	At most k exceptional triple-unit columns.
Ideal determinant filter	$J \le 2^d \le 8^k$	d nonfree special SCCs; one-sided randomization.
Deterministic width algorithm	$J(m+1)^w$	w is their implication-poset width.
Independent positive check	Polynomial	Supplied tree and actual unsigned peel order.

Here m is the number of rows, k counts exceptional columns, d counts their nonfree ordinary SCCs, and J counts the ideals of the induced reachability poset. Every ideal is a downward-closed set. Verification does not trust the Boolean table, the determinant evaluation, the random seed, or the discovery program. It checks the original finite input directly.

1.3 Where the implementation gains occur

The interval-orbit producer already used prefix-gap information in one weighted contraction path. Three independent certificate replay paths still computed the shift of each endpoint by scanning all preceding gaps. This was a real repeated operation in whole-cut verification. We supply a verifier-owned prefix implementation, preserve the exact quotient and inverse conventions, and compare against the old sums on complete certificates and independently generated finite examples.

A separate change shares internally reconstructed geometry within one composed call. On the producer side, the shared object is its own geometry. On the verifier side, it is the verifier's fresh reference reconstruction. No public prepared-object parameter or persistent cache is introduced. This distinction prevents an apparent optimization from quietly shifting the trust boundary to producer-supplied data.

1.4 Relation to Lackenby's announced target

The primary sources use two displayed quasi-polynomial exponents. Oxford's 2021 announcement states $n^{c \log n}$, whereas the detailed Oxford handout displays $n^{O((\log n)^2)}$. Both are quasi-polynomial; we do not silently identify them [13, 14]. The July 2026 preprint supplies a hierarchy algorithm, restart analysis, and compressed certification results; it does not establish the announced quasi-polynomial bound [15].

Our main theorem has an especially useful local consequence: if a supplied state has polynomial encoded size and only $O(\log n)$ exceptional rows, the randomized forest subproblem has polynomial witness-discovery time. More generally, bounded implication width gives a deterministic polynomial local algorithm, and logarithmic width gives a deterministic quasi-polynomial local algorithm. Proving that the whole hierarchy always reaches such states, or that there are only quasi-polynomially

many states of controlled size, remains separate work. A verified local improvement is a concrete component of that program, not a substitute for its global argument.

1.5 Attribution and novelty

Directed reachability, strongly connected components, matroid intersection, Cauchy–Binet, Boolean Möbius inversion, and polynomial identity testing are established tools. The standard common-basis polynomial and color sieve are explicitly treated in the algebraic matroid literature [4, 5]. The contribution here is the exact reduction of forest-compatible peeling to a bounded exceptional-row state, its Boolean determinant compilation, its integration with the native three-occurrence theorem, the implication-poset refinement, and the supplied audited implementations. Efficient lattice transforms are also classical; the relevant bound is credited to Björklund et al. [6]. We make no exhaustive literature-priority claim. All claims of progress are relative to the inspected repository and the cited mathematical inputs.

2 The finite problem and its geometric meaning

2.1 Rows, columns, and the rooted dual graph

Let D be a finite connected multigraph with distinguished root $*$. Its non-root vertices represent three-handles. Each row $r \in R$ of a nonnegative integer matrix $A = (a_{rc})$ also labels an edge of D. Rows not eligible for a top cancellation are represented by root loops; the software permits an equivalent `None` edge record. Parallel edges and loops remain distinct occurrences.

Columns represent retained one-handle cocores after a fixed lower zero/one-handle forest has been contracted. An entry a_{rc} is an *actual unsigned number of crossings* of an attaching two-handle curve with that cocore. It is not the signed exponent sum of a word. For the raw cancellation decision, all values above one may be saturated at two. This preserves whether a restriction has a column sum exactly one. Saturation must not be used for an unrelated parity computation.

Deleting the rows of a spanning tree T models the source-certified top cancellations. A private-column peel subsequently removes a row r when some column has exactly one total remaining occurrence, located on that row. It removes that pivot column as well; all other entries of the remaining rows stay unchanged. A column that has already served as a private pivot cannot later matter to an active row, because it had no other live owner when removed.

Definition 2.1 (Forest-compatible raw completion)**.** A pair (D, A) admits a forest-compatible raw completion when some spanning tree T of D leaves a row restriction $A|_{R\setminus T}$ that can be fully removed by private-column peeling.

This is the exact finite problem solved below. Under the source-authentication hypotheses of the preceding geometry compiler, success gives a valid topological cancellation sequence. Without those hypotheses, it is only a graph/matrix theorem.

2.2 Why stopping sets are the right obstruction

Definition 2.2 (Stopping set)**.** A nonempty row set S is stopping if, for each column c, $\sum_{r\in S} a_{rc}$ is either zero or at least two.

If all rows of a stopping set remain, no first private-column peel can remove a row from that set. Conversely, a nonempty residual of exhaustive column-leaf peeling is stopping. The predecessor establishes the stronger fact that allowing arbitrary positive singleton substitutions does not enlarge

the class of matrices admitting complete raw cancellation [19, 20]. We use that exact completeness result to focus on no-fill peeling. We do not introduce negative coefficients, free reduction, or unverified geometric isotopy.

The obstruction is restricted to the chosen calculus. A stopping set does not prove that a manifold is not a handlebody, and certainly does not prove that a knot is nontrivial. Conversely, some native stopping examples really are non-handlebodies, so not every refusal should be treated as a positive instance awaiting a clever cancellation order.

2.3 The existing polynomial regime

Call a column *inert* if it has no entry equal to one. Such a column cannot ever be used by a private-column peel, even after row deletion. Ignore it for the finite decision while retaining any corresponding geometric handle count.

For a useful column supported on a single row u, put u in a seed set Z. For a useful column supported on two rows u, v, add $u \to v$ when the entry on v is one, and add $v \to u$ when the entry on u is one. The other entry may exceed one. These directed arcs form the ordinary propagation graph P. From any deleted row set Q, ordinary peeling removes exactly the rows reachable from $Z \cup Q$.

If all useful columns are ordinary, every source strongly connected component of P without a seed must be hit by the tree. These components are disjoint. Independent representatives from them can be selected by intersecting the graphic matroid of D with a partition matroid and then extending the selected forest to a spanning tree. This is the predecessor's polynomial theorem, based on the classical independent-representatives and matroid-intersection results [1, 2, 3, 20].

2.4 Why native triple columns are structurally natural

An original native one-handle cocore comes from a complementary region of a triangular source face cut by normal arcs. Such a region meets each of the triangle's three sides in at most one gap. Each unsigned two-handle occurrence corresponds to an actual boundary gap. Therefore the total number of occurrences on the native cocore is at most three. Lower-forest contraction deletes columns, and top cancellation deletes rows; neither increases this bound. The preceding normalization report proves this with source-occurrence conventions [20].

The possible positive-entry multisets are consequently

$$(1), \quad (2), \quad (3), \quad (1,1), \quad (1,2), \quad (1,1,1).$$

Only the last type is outside ordinary propagation. It supplies three rules, each with two premises: $\{u, v\} \to w$, $\{u, w\} \to v$, and $\{v, w\} \to u$. This explains the focus on triple-unit columns. The stronger Boolean theorem also applies to wider exceptional columns, with the union of their incident rows as its parameter.

The native bound does not automatically survive arbitrary interface augmentation, handle slides, or a change of representation. A native adapter must measure the actual exported matrix and check the stated support conditions. Our source audit does so, and the generic Boolean solver treats wider columns without pretending that they satisfy the three-occurrence theorem.

2.5 Size and success conventions

The input size N includes explicit row and edge records and all integer bit lengths. The number of rows is m, the graphic rank is $r = |V(D)| - 1$, the number of exceptional columns is k, and their

incident-row union is U, with $t = |U|$. The source graph is checked for connectivity before arrays indexed by its vertex namespace are allocated.

A positive certificate consists of a spanning tree and a complete sequence of row/column pivot pairs. The unchanged predecessor verifier reads the original sparse matrix, checks the tree, and checks each actual column sum. Randomized zero tests and resource limits are inconclusive. The deterministic fallback can conclude exhaustive finite-model infeasibility; the present implementation does not export a compact negative certificate for that entire search. This distinction is recorded in its output schema.

3 A geometric parameter and an optimal lower forest

The raw arity statement in the preceding report is restated here with a source-level proof, followed by two refinements useful for choosing the lower forest [20]. This section distinguishes lower zero/one-handle contraction from the top two/three-handle forest problem.

3.1 Why raw source cores have column arity at most three

The three-row condition of the finite matrix solver has a direct geometric source. The following statement concerns the raw source core before any bundle-attachment rows are added. It counts geometric occurrences, including repeated incidences caused by face identifications, rather than signed intersection numbers or freely reduced words.

Proposition 3.1 (Triangular-face arity)**.** *In the maintained source parallelity decomposition of a finite triangulation, the total unsigned number of raw core two-handle belt crossings through any fixed core one-handle is at most three. It is at most two for a corner one-handle. These bounds persist after contraction of any lower spanning forest and deletion of any top-forest rows.*

Consequently the source raw matrix satisfies the three-row hypothesis: every column containing a unit entry has support in at most three rows. If the support has size three, all three entries are one and the column comes from the central region of an internally paired triangular face. For a triangulation with T tetrahedra, the number k of such exceptional columns satisfies $k \leq 2T$.

Proof. A core one-handle is dual to one region of an internally paired triangular face. The source decomposition retains its central region and, for each nonempty normal-arc corner family, the triangular region closest to that original face vertex. Every other face region is a parallelity strip. A central region meets each of the face's three original edges in at most one interval. A retained corner region meets at most the two original edges incident with its corner.

The raw core two-handles are dual to extreme gaps of original triangulation edges: one gap if the edge weight is zero and two extreme gaps otherwise. A crossing through the belt disc of the chosen core one-handle is precisely one incidence of such an extreme gap with the chosen face region. Each original face-edge side contributes at most one such incidence. In particular, a positive-weight edge cannot contribute both extreme gaps to the same face region; these are separated by the normal intersection points. For a zero-weight edge there is only one gap. Pairing the two face copies identifies their side incidences, so it does not double the count. Global identifications may make two sides incident with the same two-handle, but these remain two occurrences among the same three (or two) local sides. The unsigned total is therefore bounded as asserted.

Deleting forest columns or top-forest rows only removes incidences. Saturating an entry at two also cannot increase its column sum. Three positive entries in a column whose sum is at most three must all be one. There is at most one central-region one-handle for each internal face pair, and at most $2T$ internal face pairs in a T-tetrahedron triangulation. □

A useful refinement detects which central regions can contribute. Write c_u, c_v, c_w for the three nonnegative corner normal-arc counts in a paired face, using the matching equations to identify the counts on its two copies. The normal weight on its edge uv is $c_u + c_v$, while the central face gap has rank c_u from u. This gap is extreme exactly when $c_u = 0$ or $c_v = 0$. For a central region to contribute three raw occurrences, this must hold on all three face edges. Thus at most one of c_u, c_v, c_w may be positive. Boundary-edge exclusions or global row identifications may lower the arity further. In particular,

$$k \le \#\{\text{internal face pairs with at most one positive corner-arc family}\},$$

even before subtracting central columns chosen in the lower forest. This refinement depends on the normal-coordinate zero pattern, not on the binary lengths of its positive multiplicities.

The proposition does not apply automatically after arbitrary annular bundle rows or additional geometric relations are appended to the matrix. Nor does $k \le 2T$ by itself imply a quasipolynomial bound in diagram size. Its use is to establish a natural, source-geometric domain for the parameterized solver, while leaving the required small-k or hierarchy-wide bounds explicit.

Proposition 3.2 (An optimal lower forest for the exceptional-column parameter)**.** *Form the raw* $0/1$*-handle multigraph* G *before lower forest contraction, and let* E_3 *be its edges whose raw incidence columns contain three distinct unit rows. For a spanning forest* F *of* G*, let* $k(F)$ *be the number of such columns remaining after lower contraction, before any top-row deletion, interface augmentation, or additional geometric move. Then*

$$\kappa := \min_F k(F) = |E_3| - r_G(E_3),$$

where r_G *is the graphic-matroid rank. Equivalently,* κ *is the cycle nullity of the spanning subgraph with edge set* E_3*, counting loops and parallel edges in the usual multigraph sense. A minimum-parameter forest is constructed by processing* E_3 *first in a greedy spanning-forest algorithm, then extending with all other edges.*

Proof. Lower contraction deletes exactly the columns in F and changes no incidence in a retained column, so $k(F) = |E_3 \setminus F|$. Since $F \cap E_3$ is a forest, it has at most $r_G(E_3)$ edges. Conversely, any maximal forest in the subgraph E_3 has $r_G(E_3)$ edges and extends to a spanning forest of G. Extension never removes a chosen edge. This proves both the equality and the constructive claim, including disconnected graphs, loops, and parallel edges. □

The optimization concerns the parameter, not recognition success. Different lower forests can leave different positive-peeling problems, and a forest minimizing k need not preserve a success available for some other lower forest. A certificate produced with the chosen forest remains independently checkable; a refusal retains its restricted meaning. Thus the guaranteed $8^k\,\mathrm{poly}(N)$ finite solver can be run with the smallest available raw-core parameter κ, while exhaustive completeness over all choices of lower forest remains a separate question.

4 A parameterized extension through the triple-unit barrier

The two-row theorem in the preceding incoming report leaves exactly one native column type unresolved: three distinct rows, each of multiplicity one. The following result makes the number of these exceptional columns, rather than the number of three-handles, the parameter of an exact search. It applies to the finite forest-compatible raw-completion problem. It neither asserts that all native interfaces satisfy its hypotheses nor changes the underlying geometric certificate class.

Let D be a connected rooted multigraph, with a row set R indexing its edges, and let $A \in \mathbb{N}^{R\times C}$ be the unsigned occurrence matrix. Ineligible top-cancellation rows are loops at the root. A column is inert when it has no entry equal to one. Suppose that every non-inert column either has support of size at most two or belongs to a distinguished set E of k columns of type $(1, 1, 1)$. We call all columns outside E ordinary. Counts exceeding one can be saturated at two for the finite peeling decision, although not for source-parity checks.

Write $m = |R|$ and $r = |V(D)| - 1$. Let N denote the total explicit encoded input length, including integer bit lengths. The assertions below concern this input length; translating them into a bound in the original diagram size requires a separate bound on N and k.

4.1 Separating ordinary reachability from exceptional activations

Construct the directed ordinary implication graph P on R. An ordinary singleton unit column makes its row a seed, and the set of all such seeds is Z. An ordinary two-row column supported on $\{u, v\}$ contributes $v \to u$ when its entry on u is one and $u \to v$ when its entry on v is one. Inert columns contribute nothing. Let $\mathrm{Reach}(X)$ be reachability from X in P, including paths of length zero. For a row x, put

$$\mathrm{Anc}(x) = \{y \in R : x \in \mathrm{Reach}(\{y\})\}.$$

Ordinary peeling from a set of already removed rows Q removes exactly $\mathrm{Reach}(Z \cup Q)$. This is the incoming reachability lemma; it applies unchanged when exceptional columns are simply not used.

An *exceptional schedule* consists of distinct columns $c_1, \dots, c_\ell \in E$, where $0 \le \ell \le k$, and a designated head $h_i \in \mathrm{supp}(c_i)$ for each i. Define

$$U_i = \mathrm{supp}(c_i) \setminus \{h_i\}, \qquad H_i = \{h_1, \dots, h_i\}, \qquad H_0 = \varnothing.$$

Thus $|U_i| = 2$. The schedule is intended to use c_i only after both rows in U_i have gone. If h_i has already gone, its scheduled activation can be skipped; retaining such redundant schedules simplifies enumeration without making the criterion unsound.

For every $x \in U_i$ that is not in $\mathrm{Reach}(Z \cup H_{i-1})$, introduce the requirement

$$T \cap \mathrm{Anc}(x) \neq \varnothing. \tag{4.1}$$

List these requirements, retaining or merging duplicates, as $T \cap W_j \neq \varnothing$ for $1 \le j \le p$. There are at most $p \le 2\ell \le 2k$ of them. Let $S_1, \dots, S_q$ be the source strongly connected components of P that contain no row of $Z \cup H_\ell$. These groups are pairwise disjoint.

Theorem 4.1 (Exact schedule reduction)**.** *There is a forest-compatible raw completion if and only if some exceptional schedule admits a spanning tree T of D satisfying*

$$T \cap S_a \neq \varnothing \quad (1 \le a \le q), \qquad T \cap W_j \neq \varnothing \quad (1 \le j \le p). \tag{4.2}$$

The source components and predecessor sets are computed in the fixed ordinary implication graph; no two-premise rule is replaced by an individual directed arc.

Proof. Suppose first that T leaves a completely peelable row restriction. Choose a complete unit-column peel order, which exists by the incoming raw-pivot completeness theorem. Record, in that order, precisely the exceptional columns actually used as pivots, and record each pivot's row as its head. A column is used at most once. Before the ith recorded pivot, every removed row belongs to $\mathrm{Reach}(Z \cup T \cup H_{i-1})$: this follows by induction from the exact ordinary reachability model. Consequently both prerequisites in U_i belong to that set. Since reachability distributes over unions,

a prerequisite not reachable from $Z \cup H_{i-1}$ must be reachable from some member of T, which is exactly (4.1). At the end every row lies in $\mathrm{Reach}(Z \cup T \cup H_\ell)$. A source component can only be reached from a vertex already in that component. Therefore every source component disjoint from $Z \cup H_\ell$ meets T, as required.

Conversely, assume (4.2). Delete the rows of T. Close under ordinary peeling. At step i, the induction hypothesis says that every row of $\mathrm{Reach}(Z \cup T \cup H_{i-1})$ has been removed. Each prerequisite in U_i is either reachable from $Z \cup H_{i-1}$ or is reachable from a tree row by the associated requirement $T \cap W_j \neq \varnothing$. It has therefore been removed. If h_i remains, the exceptional column c_i has total one on the live rows and can peel h_i. It has not previously been used because the scheduled columns are distinct and ordinary closure does not use them. If h_i is already absent, skip the pivot. Close under ordinary peeling again. This establishes the induction for the next step.

Every component of the condensation DAG of P is reachable from a source component. Each source component contains a row of $Z \cup H_\ell$ or, by hypothesis, a row of T. Hence the final ordinary closure is all of R. The constructed sequence completely peels $R \setminus T$. □

A head not in a source component does not seed that source component. In particular the test for S_a must use actual membership in the source SCC, not merely downstream reachability from it. Similarly, H_{i-1}, not the final head set, is used for the ith prerequisite.

4.2 A deterministic algorithm with no dual-rank exponent

Theorem 4.2 (Deterministic exceptional-column algorithm)**.** *Under the stated hypotheses, forest-compatible raw completion is decidable in*

$$3^k k!\,(m+1)^{2k}\,\mathrm{poly}(N)$$

time and polynomial space apart from the current schedule and certificate. A positive output is a spanning tree and a unit-column peel order. The exponent hidden in the polynomial factor is independent of k and r.

Proof. Enumerate every exceptional schedule. Their number is

$$\sum_{\ell=0}^{k} \frac{k!}{(k-\ell)!} 3^\ell = 3^k k! \sum_{t=0}^{k} \frac{1}{3^t t!} < e^{1/3} 3^k k!.$$

For a fixed schedule, enumerate a representative $e_j \in W_j$ for each of its $p \leq 2k$ arbitrary requirements. Repetitions are permitted; let $Q = \{e_1, \ldots, e_p\}$. Reject the choice if Q is not a graphic forest, including when it contains an ineligible row or a loop. For every source group already met by Q, discard the group. Contract Q in the graphic matroid. The remaining source groups are still pairwise disjoint. Use graphic–partition matroid intersection to choose one representative of each remaining group, independent after this contraction. The incoming Rado/augmenting-path solver supplies precisely this operation. If it succeeds, the representatives together with Q form a forest that meets all required sets. Extend it to a spanning tree; adding edges cannot destroy a hitting condition.

Conversely, if a tree satisfying (4.2) exists, choose e_j from its intersection with W_j. The resulting Q is a forest. Choosing one tree edge from every source group not already hit gives an independent representative set after contraction, so matroid intersection succeeds on this enumerated choice. Thus no feasible schedule is missed. There are at most $(m+1)^{2k}$ representative choices. All graph, closure, contraction, and matroid-intersection operations have polynomial cost. Finally recompute and independently verify the peel order on the actual completed tree; a schedule alone is not the final geometric certificate. □

The time bound is an XP bound in k, rather than a fixed-parameter bound. It is nevertheless independent of an exponential dependence on the number r of dual-tree edges. If $N = n^{O(1)}$ and $k = O(\log n)$, it gives the local bound $\exp(O((\log n)^2))$. This is not an end-to-end unknot recognition bound.

4.3 A determinant sieve for the few overlapping requirements

We give a self-contained algebraic lemma because it cleanly removes the factor m^{2k} with one-sided randomization. Its ingredients are Cauchy–Binet, inclusion–exclusion, and polynomial identity testing, all standard in algebraic matroid algorithms; compare Harvey's algebraic matroid-intersection work and Eiben–Koana–Wahlström's treatment of enumerating polynomials. The application and schedule reduction above are the contribution here.

Lemma 4.3 (A spanning tree with partition and covering requirements)**.** *Fix $0 < \varepsilon < 1$. Let D be a connected graph of rank r, let $S_1, \dots, S_q$ be pairwise disjoint edge groups, and let $W_1, \dots, W_p$ be arbitrary edge sets. There is a randomized algorithm, with time $2^p \operatorname{poly}(N, \log(1/\varepsilon))$, that returns a checked spanning tree meeting all groups and sets whenever it finds one. It never returns an incorrect tree. If a feasible tree exists, it returns one with probability at least $1 - \varepsilon$. If no feasible tree exists, it never reports success. It uses polynomial space.*

Proof. If $q > r$, feasibility is impossible. Otherwise make one filler copy e^0 of every edge e and, for $e \in S_a$, one representative copy e^a. Because the groups are disjoint there are at most $M \le 2m$ copies. Choose a prime field $\mathbb{F}_P$ with

$$P > \max\{M, 2r\}.$$

Choose distinct field elements λ_e for the filler copies. Let A be the reduced oriented incidence matrix of D, of size $r \times M$, repeated on the copies. A set of r copies is a basis of A exactly when its underlying edges form a spanning tree, with no edge repeated.

Construct an $r \times M$ matrix B with a top block of q rows and a bottom block of $r - q$ rows. The representative copy e^a has column $(\mathbf{e}_a, 0)$; the filler copy e^0 has column

$$(0, 1, \lambda_e, \dots, \lambda_e^{r-q-1})^{\mathsf{T}}.$$

When $r = q$, this filler column is the zero column. A set of r columns is a basis of B exactly when it contains one representative of each S_a and $r - q$ distinct fillers. This follows from the block structure and the nonzero Vandermonde determinant for distinct λ_e. Consequently common bases of A and B encode exactly the spanning trees meeting all the disjoint groups, together with a chosen representative edge in each group. Different encodings are harmless and receive distinct variables below.

Assign an independent variable z_f to each copy f. For $J \subseteq \{1, \dots, p\}$, retain only copies whose underlying edge belongs to none of the sets W_j with $j \in J$, and let $D_J(z)$ be the diagonal matrix with entry z_f on retained copies and zero elsewhere. Define

$$\Phi(z) = \sum_{J \subseteq [p]} (-1)^{|J|} \det(A D_J(z) B^{\mathsf{T}}). \tag{4.3}$$

Cauchy–Binet expands each determinant into

$$\sum_{F:\, |F| = r} \det(A_F) \det(B_F) \prod_{f \in F} z_f.$$

For a common basis F, its coefficient in (4.3) is multiplied by $\sum_{J\subseteq\{j:F\cap W_j=\varnothing\}}(-1)^{|J|}$. This multiplier is one exactly when every W_j is met, and zero otherwise. Here $F\cap W_j$ means that some copy in F has its underlying edge in W_j. The remaining coefficients are nonzero: graphic basis determinants are ±1, and the required Vandermonde determinants are nonzero. Every distinct set F has a distinct monomial because variables belong to copies, not merely to original edges. Thus cancellation between different encodings cannot make Φ identically zero. We have proved

$$\Phi \not\equiv 0 \quad\Longleftrightarrow\quad \text{a feasible spanning tree exists.}$$

The polynomial Φ is multilinear and homogeneous of degree r. Sample the z_f independently and uniformly in $\mathbb{F}_P$. The Schwartz–Zippel bound [11, 12] gives $\Pr[\Phi(z)=0 \mid \Phi\not\equiv 0]\le r/P<1/2$. Repeat independently $b=\lceil\log_2(1/\varepsilon)\rceil$ times (or use one trial when $b=0$); the probability that all trials miss is at most ε. A nonzero evaluation is conclusive evidence that a feasible basis exists. Each evaluation requires 2^p determinants of order r and polynomial additional work. Bertrand's postulate supplies such a prime below twice a polynomial threshold in $M+r$, and deterministic trial division suffices to find it in polynomial time. Arithmetic has bit cost polynomial in $\log N$, and repetition contributes a factor $O(1+\log(1/\varepsilon))$.

For completeness, a positive evaluation yields a witness without any additional probabilistic decisions. Keep the sampled values fixed. Repeatedly scan the current set U of copies, deleting f whenever the restricted value $\Phi_{U\setminus\{f\}}(z)$ remains nonzero. Stop after an entire scan deletes nothing. At most M scans that delete something occur, so at most $O(M^2)$ evaluations are needed. The final restricted polynomial has value $a\neq 0$, and deleting any one of its variables gives zero. Multilinearity therefore gives

$$z_f\frac{\partial\Phi_U}{\partial z_f}(z)=a \qquad (f\in U).$$

Euler's identity for a homogeneous polynomial of degree r gives

$$|U|a=\sum_{f\in U} z_f\frac{\partial\Phi_U}{\partial z_f}(z)=ra.$$

Since $P>M\ge|U|$ and $r\le M$, and $a\neq 0$, this forces $|U|=r$. Its only possible degree-r squarefree monomial is the product of all remaining variables. Its nonzero value proves that U is a common basis meeting all the requirements. Decode the underlying tree and check its connectivity and every hitting condition directly over the integers. This extraction uses no fresh random oracle, hence incurs no additional failure probability.

All restrictions and determinants can be processed sequentially, so the space is polynomial. If the initial value is zero the algorithm reports that no witness was found; it does not manufacture a negative certificate. □

Prime selection and an optional simpler error convention. For fixed error, choose a deterministic prime P larger than $\max(M,2r)$; repeating independent trials b times makes the miss probability at most 2^{-b}. A prime in a polynomial interval can be found by deterministic trial division, already sufficient for the stated polynomial input-size bound. This avoids assuming a particular fast prime-generation implementation. The extraction scan needs a full final pass with no deletion; a single pass need not be inclusion-minimal because later deletions can change earlier cancellation in the field.

Composing this covering lemma with schedule enumeration gives the valid but weaker randomized factor $O(12^k k!)$ in polynomial space. The reference implementation retains this route for comparison. The next two sections remove the factorial schedule enumeration entirely.

5 Compressing all exceptional schedules into a Boolean determinant filter

The schedule method exposes a deterministic exact algorithm, but its factorial enumeration is unnecessary for randomized discovery. Every exceptional column can change only one of its incident rows. Keeping track of those rows captures all nonlinear activation, while ordinary rows are handled by graph reachability and the fixed source-component constraints. This gives a single-exponential parameter bound.

Let P and Z be the ordinary implication graph and ordinary seed set, and let

$$U = \bigcup_{c \in E} \operatorname{supp}(c), \qquad t = |U| \le 3k.$$

Use $\mathrm{Cl}(X)$ for the closure of $X \cup Z$ under ordinary reachability and all exceptional rules. For a triple $\{u, v, w\}$ these rules are $\{u, v\} \to w$, $\{u, w\} \to v$, and $\{v, w\} \to u$. Equivalently, $\mathrm{Cl}(X)$ consists of the rows removed by peeling after initially deleting X. All rules concern removed rows.

Define a monotone Boolean function on subsets of the special rows by

$$f(A) = \mathbf{1}\{U \subseteq \mathrm{Cl}(A)\}, \qquad A \subseteq U. \tag{5.1}$$

The seeds Z are always included in this definition. They will not be encoded as colors that a tree is required to hit. In particular f may already be one on the empty set.

Let $S_1, \dots, S_q$ now denote the source SCCs of the ordinary graph P that are disjoint from $Z \cup U$. These groups are fixed for the entire algorithm, independently of the tree and the Boolean input A. For a spanning tree T, define its special influence set by

$$H(T) = \mathrm{Reach}_P(T) \cap U = \{u \in U : T \cap \mathrm{Anc}(u) \neq \varnothing\}. \tag{5.2}$$

Only influence from T is included here. Influence from the free seeds is already accounted for in f.

Lemma 5.1 (Exact special-row criterion)**.** *A spanning tree T leaves a completely peelable matrix if and only if*

$$T \cap S_a \neq \varnothing \quad (1 \le a \le q), \qquad f(H(T)) = 1. \tag{5.3}$$

Proof. First we prove that the special rows activated from T are exactly those activated from $H(T)$. Before any exceptional rule fires, their special sets are respectively

$$\mathrm{Reach}_P(Z \cup T) \cap U \quad \text{and} \quad \mathrm{Reach}_P(Z \cup H(T)) \cap U.$$

These sets agree. Every member of $H(T)$ is reachable from T; conversely every special row reachable from T is already a member of $H(T)$. If the special sets agree at some stage, precisely the same exceptional rules are enabled, because every premise and head of every such rule lies in U. Their new heads also lie in U. Ordinary closure from a common set of new special heads has the same effect on special rows. Induction over the finite monotone closure process proves

$$\mathrm{Cl}(T) \cap U = \mathrm{Cl}(H(T)) \cap U. \tag{5.4}$$

There is no need to retain the initially activated ordinary rows outside U: their only future effect on special rows has already been included in $H(T)$.

If all rows peel, (5.4) gives $f(H(T)) = 1$. A source SCC disjoint from $Z \cup U$ cannot receive an ordinary implication from another SCC and cannot contain an exceptional head or a seed. It therefore must contain a deleted tree row. This proves necessity.

Conversely suppose (5.3). All special rows become active by (5.4). Every source SCC of P then contains an active row: a seed in Z, a special row in U, or a tree row by the hitting condition. Every SCC of a finite condensation DAG is reachable from a source SCC. Ordinary closure therefore activates all rows. This is complete peeling. □

5.1 Computing the activation table without expanded words

The table of f can be computed in $2^t \operatorname{poly}(N)$ time. Precompute ordinary reachability between special rows and the free mask $\operatorname{Reach}_P(Z) \cap U$. For each initial mask A, repeatedly add the ordinary successors of active special rows and every exceptional head whose prerequisites are active. At most t strict mask enlargements can occur. The resulting mask is all of U exactly when $f(A) = 1$. This is a finite monotone closure computation on bit masks, not an exponential expansion in normal-surface weight.

For a more economical parameter, let d be the number of ordinary SCCs that meet U. Two rows of one SCC have identical ordinary ancestor sets, and one becomes active ordinarily whenever the other does. Hence one color per such SCC suffices. Choose one representative of each SCC, interpret an initial color mask as activating its representatives, and use the same closure test on all special rows. Thus all statements below hold with t replaced by

$$d \le t \le 3k. \tag{5.5}$$

For clarity the algebraic proof is written with the individual-row set U; replacing it by SCC colors changes none of its identities.

5.2 An arbitrary Boolean filter on common bases

Use the copied graphic-incidence/Vandermonde construction from Lemma 4.3, with the fixed groups $S_1, \dots, S_q$ from Lemma 5.1. If $q > r$, no tree can hit all these disjoint groups. Otherwise common bases of the two represented matroids correspond exactly to spanning trees that hit all these groups. There are at most $M \le 2m$ copies, each with its own variable z_e.

Assign to a copy of a row e the color mask

$$C(e) = \{u \in U : e \in \operatorname{Anc}(u)\}.$$

The union of the masks in a copied spanning-tree basis is exactly $H(T)$. For $A \subseteq U$, let

$$P_A(z) = \det(A_{\mathrm{gr}} D_A(z) B^{\mathsf{T}}),$$

where $D_A(z)$ retains z_e if $C(e) \cap A = \varnothing$, and sets that entry to zero otherwise. Thus a common basis F occurs in P_A exactly when its hit mask $H(F)$ avoids A.

Define

$$g(K) = f(U \setminus K), \qquad c_A = \sum_{B \subseteq A} (-1)^{|A \setminus B|} g(B), \tag{5.6}$$

and form the polynomial

$$Q(z) = \sum_{A \subseteq U} c_A P_A(z). \tag{5.7}$$

Theorem 5.2 (Boolean common-basis sieve)**.** *The polynomial Q is multilinear and homogeneous of degree r. Its monomials with nonzero coefficient encode exactly the copied spanning-tree bases whose underlying trees satisfy (5.3). In particular*

$$Q \not\equiv 0 \quad \Longleftrightarrow \quad \textit{forest-compatible raw completion exists.}$$

This holds over any field for which the two matrix representations are valid. The Boolean function need not be monotone for the algebraic identity; monotonicity enters only its interpretation as activation.

Proof. Cauchy–Binet associates a common basis F with the distinct monomial $\prod_{e\in F} z_e$ and its nonzero coefficient $\det((A_{\mathrm{gr}})_F)\det(B_F)$. It occurs in P_A precisely when $A \subseteq U \setminus H(F)$. Its coefficient in Q is therefore its original coefficient multiplied by

$$\sum_{A\subseteq U\setminus H(F)} c_A = g(U \setminus H(F)) = f(H(F)).$$

The middle identity is subset-lattice Möbius inversion. To verify it directly, substitute (5.6), interchange the two finite sums, and use $\sum_{A:\, B\subseteq A\subseteq K}(-1)^{|A\setminus B|} = 0$ unless $B = K$, in which case it is one. Thus the multiplier is exactly zero or one. The identity holds over the integers and after reduction in any characteristic. Large or negative intermediate c_A do not introduce an additional characteristic restriction.

Distinct bases use distinct copy variables, so different encodings do not share a monomial. Every surviving basis coefficient remains nonzero. The exact special-row criterion now proves the final equivalence. Multilinearity and homogeneity follow directly from the degree-r Cauchy–Binet monomials. □

5.3 Single-exponential discovery with checked witnesses

Theorem 5.3 (Activation-dimension algorithm)**.** *Let d be the number of ordinary SCCs incident with exceptional columns. For every $0 < \varepsilon < 1$, there is a randomized algorithm with time*

$$2^d \operatorname{poly}(N, \log(1/\varepsilon)) \;\le\; 8^k \operatorname{poly}(N, \log(1/\varepsilon)) \tag{5.8}$$

that finds a spanning tree and a verified unit-column peel order with probability at least $1-\varepsilon$ whenever a completion exists. It never reports an incorrect positive certificate. It uses $O(2^d \log N + \operatorname{poly}(N))$ bits of space. There is also a polynomial-space variant with time

$$3^d \operatorname{poly}(N, \log(1/\varepsilon)).$$

Proof. Compute ordinary SCCs, ancestor masks, and the truth table on the d SCC colors. Choose a prime field with characteristic $P > \max(M, 2r)$, of polynomial magnitude in N. The represented graphic matroid and the filler Vandermonde matroid are valid over this field. Compute g and its subset Möbius transform c by the standard in-place recurrence

$$c[A] \leftarrow c[A] - c[A \setminus \{i\}] \quad \text{for each color } i \text{ and each } A \ni i.$$

There are $O(d2^d)$ field operations. All coefficients may be computed modulo P; their integer identities have already been proved.

An evaluation of Q uses 2^d determinants of order r and polynomial additional work. If completion exists, Q is a nonzero degree-r polynomial, so a uniform random assignment misses with probability at most $r/P < 1/2$. With $b = \max\{1, \lceil \log_2(1/\varepsilon)\rceil\}$ independent trials, the probability of missing every time is at most ε. A nonzero evaluation guarantees existence without qualification.

Use the deterministic repeated-deletion extraction from Lemma 4.3, now with Q in place of Φ. All of its hypotheses hold: Q is homogeneous and multilinear of degree r, and $P > M$. At most $O(M^2)$ further evaluations produce exactly r copies forming a common basis with $f(H) = 1$. Decode its tree and run the independent unit-column peeling checker on the original matrix. Lemma 5.1

proves that every row is removed. No random choices are used during extraction, so the error bound remains that of the initial identity tests.

Storing f, g, and c costs $O(2^d \log N)$ bits when values are stored modulo P, in addition to the polynomial graph and matrix storage. For polynomial space, do not store these tables. Compute each c_A directly from its subsets, evaluating f by closure as needed. Summed over all A, the number of truth-value computations is

$$\sum_{A \subseteq [d]} 2^{|A|} = 3^d.$$

Every determinant, coefficient, and restriction can then be processed sequentially. The repeated extraction calls multiply the time only by a polynomial factor. □

The randomized algorithm has one-sided misses. A production interface can report `SUCCESS` only after exact replay, and return `INCONCLUSIVE` when the tested assignments are all zero. An exact negative decision may use the deterministic schedule solver, an existing stopping-set certificate, or exhaustive search. Probabilistic failure to find a witness is not a topological obstruction.

Corollary 5.4 (Larger exceptional supports)**.** *The activation-dimension theorem remains valid if exceptional active columns have arbitrary finite support and arbitrary positive counts, provided that U contains every row incident with them. For a unit entry on row h, its rule requires every other support row to have been removed. Columns with no unit entry remain inert. The running time is $2^d \operatorname{poly}(N, \log(1/\varepsilon))$, where d counts the ordinary SCCs meeting this larger exceptional-row set.*

Proof. Every exceptional rule still has all its premises and head in U. The exact special-row criterion and Boolean filter use no bound on rule arity. A finite closure test is polynomial in the explicitly encoded rule incidence. The triple-specific inequality $d \le 3k$ is the only part that no longer applies. □

Concrete consequence and its boundary. If current states have encoded size $N = n^{O(1)}$ and activation dimension $d = O(\log n)$, this local tree-selection problem has randomized polynomial-time certificate discovery. For $d = O((\log n)^a)$ it has quasi-polynomial local time. No such dimension bound is established for all states of a general unknot hierarchy. The theorem also leaves intact fixed-interface stopping obstructions: it decides the same restricted geometric move class more efficiently, and does not make that class complete for handlebody recognition.

6 Restricting the sieve to order ideals of the implication poset

There is further structure in the possible color masks: ordinary reachability forces them to be upward closed. The full Boolean cube therefore contains states that no spanning tree can induce. Removing these states improves both the parameter and the practical table size.

Collapse ordinary SCCs meeting the exceptional supports to colors, and discard the colors already reachable from Z. Let $\mathcal{P}$ be the partial order on the remaining d colors in which

$$u \preceq v \quad \Longleftrightarrow \quad \text{an ordinary directed path leads from color } u \text{ to color } v.$$

Distinct SCCs make this relation antisymmetric. Reachability is taken in the full ordinary graph, including intermediate rows outside the special set. Let $\mathcal{I}(\mathcal{P})$ be the set of order ideals, and write

$$J = |\mathcal{I}(\mathcal{P})|.$$

Here an ideal K satisfies $v \in K$ and $u \preceq v \Rightarrow u \in K$.

Lemma 6.1 (Only ideals can be missed)**.** *The color set $H(T)$ ordinarily activated by a tree is upward closed. Consequently $K(T) = \mathcal{P} \setminus H(T)$ belongs to $\mathcal{I}(\mathcal{P})$. The same conclusion holds for the union of the color masks of any subset of edge copies, whether or not that subset is a spanning-tree basis.*

Proof. If an edge row reaches u and $u \preceq v$, transitivity implies that it reaches v. Each individual edge mask is therefore upward closed, and so is a union of such masks. Its complement is an order ideal. Discarding free colors does not change this statement for the induced partial order on the remaining colors. □

Retain the Boolean function f from the preceding section, with its free seeds included, but evaluate it only on the upward-closed masks $\mathcal{P} \setminus K$. Define

$$g(K) = f(\mathcal{P} \setminus K) \qquad (K \in \mathcal{I}(\mathcal{P})).$$

For an ideal A, let $\max(A)$ be its set of maximal elements and put

$$c(A) = \sum_{S \subseteq \max(A)} (-1)^{|S|} g(A \setminus S). \tag{6.1}$$

Every set $A \setminus S$ occurring here is again an ideal.

Lemma 6.2 (Explicit inversion on the ideal lattice)**.** *For every ideal K,*

$$\sum_{A \in \mathcal{I}(\mathcal{P}),\, A \subseteq K} c(A) = g(K). \tag{6.2}$$

Moreover the full coefficient table can be computed in $O(dJ)$ additions and subtractions, without enumerating the subsets in (6.1) *separately.*

Proof. Fix an ideal $B \subseteq K$ and collect the coefficient of $g(B)$ in the left side of (6.2). A contributing term is an ideal $A = B \cup S$ with $S \subseteq \max(A)$. This is equivalent to taking S to be an arbitrary subset of the minimal elements of $K \setminus B$. Indeed, if $s \in A \setminus B$ is maximal in A, all its strict predecessors lie in B: any predecessor outside B would lie in $A \setminus B$ and would not be maximal. Hence s is minimal outside B. Conversely, minimal elements outside an ideal B are pairwise incomparable and have every predecessor in B; adjoining any subset of them gives an ideal in which those new elements are maximal. Thus the coefficient of $g(B)$ is

$$\sum_{S \subseteq \min(K \setminus B)} (-1)^{|S|}.$$

It is one if $B = K$ and zero otherwise, since a nonempty finite poset has a minimal element. This proves the inversion identity over the integers, and hence over every field.

For the efficient transform, choose a linear extension of the partial order, with predecessors before successors. Initialize $c(A) = g(A)$ for all ideals. Process colors v in *reverse* linear-extension order. For every ideal A for which $v \in A$ and $A \setminus \{v\}$ is an ideal, perform

$$c(A) \leftarrow c(A) - c(A \setminus \{v\}). \tag{6.3}$$

The set-removal condition says exactly that v is maximal in A. To expand the coefficient eventually obtained for a fixed A, each subtraction removes a maximal element and contributes one minus sign. Any predecessor of a removed element occurs later in the processing order. It has therefore not

yet been processed in the source value when that subtraction takes place. Such a newly maximal predecessor cannot be removed along the resulting expansion path. Consequently the only elements removed are members of the original $\max(A)$, and any subset of them is removed exactly once, in reverse linear-extension order. This gives exactly (6.1). There are at most dJ updates.

The order matters. For a chain $a \prec b$, reverse processing yields $c(\{a,b\}) = g(\{a,b\}) - g(\{a\})$. Processing predecessors first would incorrectly introduce an extra $g(\varnothing)$ term in this coefficient. □

The transform is the Möbius inversion for the distributive lattice of order ideals; the argument above establishes the needed formula directly. Efficient lattice zeta and inverse transforms are established tools [6, 7]. No assumption about a new matroid of peelable sets is being made.

For each ideal A, let P_A be the common-basis determinant that bans copies whose color masks meet A, as in the preceding section. Form

$$Q_{\mathcal{I}}(z) = \sum_{A \in \mathcal{I}(\mathcal{P})} c(A) P_A(z). \tag{6.4}$$

Theorem 6.3 (Ideal-state determinant sieve)**.** *The nonzero monomials of $Q_{\mathcal{I}}$ encode exactly the copied spanning-tree bases leaving a completely peelable matrix. In particular, forest-compatible raw-completion witnesses can be found, with one-sided miss probability at most ε, in*

$$J \operatorname{poly}(N, \log(1/\varepsilon)) \tag{6.5}$$

time and $J \operatorname{poly}(N)$ space. Every success is checked by replaying a spanning tree and a unit-column peel order in the original matrix. The theorem applies to arbitrary exceptional supports through their finite Horn rules.

Proof. Fix a copied common basis F, with underlying spanning tree T. Its monomial occurs in P_A exactly when $A \subseteq K(T)$. By Lemma 6.1, $K(T)$ is an ideal. Therefore its coefficient in (6.4) is its original nonzero basis coefficient multiplied by

$$\sum_{A \in \mathcal{I}(\mathcal{P}),\, A \subseteq K(T)} c(A) = g(K(T)) = f(H(T)).$$

This is zero or one by Lemma 6.2. The fixed source groups are already enforced by the represented partition/filler matroid, so the exact special-row criterion identifies precisely the successful trees. Different copied bases still have different monomials. The polynomial is homogeneous and multilinear of degree r.

Enumerate ideals by a depth-first traversal of a linear extension. At the step for color v, exclusion is always allowed; inclusion is allowed precisely when every predecessor of v has already been included. Each leaf is an ideal and every ideal occurs exactly once. Every prefix can be extended to a complete ideal by excluding all remaining elements, so the number of prefixes at any fixed depth is at most J. There are at most $(d+1)J$ prefixes in total. This gives $J \operatorname{poly}(N)$ enumeration time, including exact predecessor mask tests.

Compute g on these J states by the polynomial-time closure routine, and compute c by Lemma 6.2. Each evaluation of $Q_{\mathcal{I}}$ uses at most J determinants and polynomial additional work. The same prime-field identity test and repeated-deletion witness extraction as before apply, with unchanged degree and characteristic requirements. A polynomial number of determinant evaluations suffices for extraction, and independent trials contribute only the stated $\log(1/\varepsilon)$ factor. Lists of ideal masks and their coefficient values use $J \operatorname{poly}(N)$ space. □

Corollary 6.4 (Width and chain bounds)**.** *Suppose the implication poset admits a partition into $w \geq 1$ chains of lengths $n_1, \ldots, n_w$, where $\sum_i n_i = d$. Then*

$$J \leq \prod_{i=1}^{w} (n_i + 1) \leq \left(1 + \frac{d}{w}\right)^w. \tag{6.6}$$

In particular, if w is the width of the poset, a chain partition of that size exists by the classical finite chain-decomposition theorem of Dilworth. Fixed width gives randomized polynomial-time discovery for this local completion problem even when d is large. If $N = n^{O(1)}$ and $w = O(\log n)$, the local time is quasi-polynomial in n. When $d = 0$, the unique ideal is the empty set and $J = 1$.

Proof. A partition into w chains exists by Dilworth's theorem [8]. An ideal intersects each chain in a prefix, so it is uniquely determined by the w prefix lengths. This injects the ideals into $\prod_i \{0, \ldots, n_i\}$ and proves the first inequality. The second is the arithmetic–geometric mean inequality applied to the positive numbers $n_i + 1$. Since $d \leq N$, the final complexity assertions follow from Theorem 6.3. A chain partition is needed only to state or certify the bound; the ideal enumeration algorithm does not need to find a minimum-width partition. □

This refinement changes the governing parameter from the number of exceptional columns to the number of feasible ordinary activation patterns. A chain has $d + 1$ ideal states; an antichain has 2^d and recovers the full Boolean bound. Intermediate posets interpolate between these cases. An antichain example shows why a small number of states cannot be assumed merely from having compressed coordinates. No bound on J, d, or w for every state of an unknot hierarchy is established here.

Theorem 6.5 (Deterministic discovery parameterized by implication width)**.** *The same finite completion problem has a deterministic exact algorithm with time*

$$J(m+1)^w \operatorname{poly}(N) = N^{O(w+1)}$$

and polynomial space, where w is the width of the nonfree special-SCC implication poset. In particular fixed width gives deterministic polynomial time, and $w = O(\log n)$ gives deterministic quasi-polynomial local time when $N = n^{O(1)}$.

Proof. Stream the ideals K and set $A = \mathcal{P} \setminus K$. Discard the state if $f(A) = 0$. Otherwise require a spanning tree to hit every fixed mandatory source group and every set $\operatorname{Anc}(v)$ for $v \in \min(A)$. The set of minima is an antichain and has at most w elements. Enumerate one eligible edge representative for each such ancestor requirement. Reject a selection containing a cycle; contract its forest; discard the mandatory groups it has already hit; and use graphic–partition matroid intersection for the remaining disjoint groups, exactly as in Theorem 4.2. Extend a successful forest to a spanning tree and verify the original peel order.

For soundness, an upward-closed set containing every minimal element of A contains all of A. Thus the tree's ordinary influence satisfies $A \subseteq H(T)$, and monotonicity gives $f(H(T)) \geq f(A) = 1$. The fixed mandatory groups complete the exact special-row criterion. For completeness, any successful tree occurs in the case $A = H(T)$; it has a representative in each of the corresponding ancestor sets, and those representatives are contained in the tree and hence independent. The remaining group representatives can also be selected from that tree.

There are J states and at most $(m+1)^w$ representative tuples per state. The width bound on J proves the stated input-size bound. Depth-first ideal enumeration, a single closure computation, and a single representative choice can all be retained at a time. The matroid intersection computation uses polynomial space. The zero-dimensional case has one state and no ancestor requirements. □

6.1 An exact counting alternative when few source groups remain

The common-basis sieve detects a mixed-sign polynomial by randomized evaluation; distinct copy variables prevent formal cancellation between different encodings. When the number q of fixed mandatory source groups is small, these requirements can instead be included directly by inclusion–exclusion. The spanning-tree polynomial then has positive coefficients, permitting deterministic exact counting.

Theorem 6.6 (Kirchhoff counting sieve)**.** *Let q be the number of fixed mandatory source SCC groups and J the number of ideals of the nonfree special-SCC implication poset. The number of spanning trees that leave a completely peelable matrix can be computed exactly and deterministically in*

$$2^q J \operatorname{poly}(N)$$

bit operations. If this number is positive, a spanning tree and verified peel order can be extracted within the same asymptotic bound. In particular, when $q = 0$ the $J \operatorname{poly}(N)$ discovery bound is deterministic.

Proof. Use the original graph edges, with one variable x_e per edge and no representative/filler copies. Let G be the reduced oriented incidence matrix of the graph. For an ideal A and a subset $B \subseteq \{1, \dots, q\}$, ban every edge whose special color mask meets A, and every edge belonging to a mandatory group S_i with $i \in B$. Let $X_{A,B}(x)$ be the diagonal matrix that retains x_e on the remaining edges and has zero entries on banned edges. Put

$$R(x) = \sum_{A \in \mathcal{I}(\mathcal{P})} c(A) \sum_{B \subseteq [q]} (-1)^{|B|} \det\bigl(G X_{A,B}(x) G^{\mathsf{T}}\bigr).$$

Cauchy–Binet gives the spanning-tree form of the matrix–tree theorem [9]:

$$\det(GXG^{\mathsf{T}}) = \sum_{T \text{ a spanning tree}} \prod_{e \in T} x_e,$$

with the sum restricted to retained edges. A tree minor has determinant ± 1, so the product of its two identical minors is one. Thus every original spanning tree occurs with coefficient one, without competing signed representative encodings.

Inclusion–exclusion over B removes precisely the trees missing a mandatory group. The ideal-lattice inversion over A multiplies every remaining tree coefficient by $f(H(T))$, exactly as in Theorem 6.3. Consequently

$$R(x) = \sum_{T \text{ successful}} \prod_{e \in T} x_e.$$

Its value at $x_e = 1$ for every edge is the exact nonnegative integer number of successful trees.

All computations have polynomial bit lengths. Each unweighted graph minor counts at most $\binom{m}{r} \le (m+1)^r$ trees, and $|c(A)| \le 2^{|\max(A)|} \le 2^d$. Hence every partial sum in the displayed formula has bit length at most

$$O\bigl(r \log(m+1) + q + \log J + d + 1\bigr),$$

which is polynomial in the explicit input length. Integer determinants can be evaluated by fraction-free elimination in polynomial bit time [10]. There are at most $2^q J$ such determinants per counting query.

For witness extraction, scan the graph edges and remove e whenever the count on the remaining graph stays positive. If removal fails, every successful tree in the current graph contains e; later

removals cannot invalidate that necessity. One pass therefore suffices. At its end the remaining graph contains a successful tree, and every remaining edge belongs to every such tree. Its edge set is thus itself one successful spanning tree. There are at most m additional count queries. Replay unit-column peeling in the original matrix to produce the final certificate. □

This is an additional exact algorithm for a particular parameter regime, using classical matrix–tree positivity. It does not remove the factor 2^q when many independent mandatory source groups remain. In that regime, the common-basis construction handles the groups in polynomial cost, or the deterministic width algorithm may be preferable.

7 Removing repeated source construction and quadratic contraction replay

This section records two implementation changes at the pinned source revision. Both preserve every complete certificate and every native terminal decision in the tested corpus. The first avoids reconstructing the same finite source charts twice in a single composed call. The second improves a repeatedly executed substep of the independent orbit checker from quadratic to quasilinear arithmetic work in the number of current interval pairings. Neither changes the class of certificates accepted, the number of AHT reduction cycles, or the scope of the normal-cut handlebody recognizer.

7.1 What the profile actually identifies

The profiled workload is the repository's complete `marked_parallelity_research/corpus.json`, with 106 source cuts. The unmodified pinned implementation returns 92 complete handlebody censuses and 14 inconclusive outcomes; every complete certificate was replayed. A full `cProfile` run recorded approximately 545 million function calls. The cumulative times are diagnostic measurements under instrumentation, not unprofiled performance claims. The producer's static-support contraction routine accounted for 44.982 seconds; independent weighted replay accounted for 23.075 seconds; unweighted orbit verification accounted for 19.048 seconds. These nested categories overlap and must not be added. By comparison, all 184 calls to the independent source parallelity reconstruction accounted for 1.600 seconds. Thus source reuse is especially useful for small or source-dominated cuts; the main measured workload is dominated by orbit reduction and transport.

The precise avoidable checker cost is visible in three independent replay paths. To remap each of the four endpoints of every current pairing after removing static gaps, they separately computed

$$x - \sum_{[\ell,r]\in\mathcal{G},\ r<x} (r-\ell+1).$$

For k pairing rows and g gaps, this costs $O(kg)$ integer arithmetic and comparison operations. The producer's weighted row mapper already used a prefix table; the new work extends the efficient idea to the independently implemented verifier and proves the common mathematical contract explicitly.

7.2 An order quotient and its inverse

Use zero-based coordinates $U_N = \{0, \dots, N-1\}$ in this proof; the maintained one-based coordinates are obtained by translating all labels by one. Let

$$G_j = [\ell_j, r_j] \cap \mathbb{Z} \qquad (1 \le j \le g)$$

be the increasing maximal gaps in the union of the domains and ranges of a finite interval-pairing system. Consequently every pairing domain and every pairing range is disjoint from every G_j. Put

$$L_j = r_j - \ell_j + 1, \qquad S_0 = 0, \qquad S_j = \sum_{i=1}^{j} L_i, \qquad q(x) = \#\{j : r_j < x\}.$$

Theorem 7.1 (Exact contraction and prefix replay)**.** *The function*

$$\phi(x) = x - S_{q(x)}$$

is the increasing bijection from $U_N \setminus \bigcup_j G_j$ to U_{N-S_g}. For each original interval isometry, conjugation by ϕ is the interval isometry with the same orientation sign and remapped endpoints. Every deleted point is an isolated orbit, so contraction reduces the total orbit count by exactly S_g and preserves the orbit relation on all remaining points.

The increasing anchors $\alpha_j = \ell_j - S_{j-1}$ determine the inverse:

$$\phi^{-1}(y) = y + S_{\#\{j:\alpha_j \le y\}}, \qquad 0 \le y < N - S_g.$$

After the gaps have been authenticated, all endpoints of k pairing rows can be remapped in

$$O(g + k \log(g+2))$$

integer arithmetic and comparison operations and $O(g+k)$ integer words of working space. For B-bit endpoints, the elementary bit bound is $O((g + k\log(g+2))B)$.

Proof. For an undeleted point x, the number of deleted points strictly below it is exactly $S_{q(x)}$. Therefore $\phi(x)$ counts the undeleted points strictly below x, proving both monotonicity and bijectivity. On any pairing domain $[a, b]$, the deleted-prefix count is constant because a static gap cannot meet that domain. The same is true on its paired range $[c, d]$. Thus ϕ restricts to a translation on each interval. Conjugating a translation or reflection by these translations gives an interval isometry of the same sign; its endpoints are precisely $\phi(a), \phi(b), \phi(c), \phi(d)$. Hence the pairing graph restricted to retained vertices is carried isomorphically to the contracted pairing graph. A static point has no nontrivial incident pairing arrow and is an isolated orbit.

After the first $j-1$ gaps have been removed, the beginning of gap j has coordinate α_j. Inserting that gap shifts exactly the points at or after α_j. The gaps are maximal, so successive gaps are separated by at least one retained point, and their anchors are strictly increasing. Counting the gaps whose anchors do not exceed y gives the displayed inverse formula. This also covers a leading gap; a trailing gap has its anchor outside the contracted universe and is never inserted below an allowed y.

Build the ordered list (r_j) and prefix sums (S_j) in $O(g)$ operations. The strict inequality $r_j < x$ is implemented by lower-bound binary search for x in (r_j). Four searches and four subtractions per row give the stated bound. The integers never exceed the original universe size, so addition, subtraction and comparison have the claimed elementary B-bit cost. No point of a gap is expanded. □

Corollary 7.2 (Replay cost per authenticated contraction)**.** *If a contraction acts on $k \ge 1$ interval-pairing rows, its maximal static-gap list has at most $2k+1$ entries. The endpoint-remapping part of the old checker had an $O(k^2 B)$ worst-case bound; the prefix implementation has an $O(k \log(k+2)B)$ bound. More precisely, a fixed trace with contraction events indexed by i changes its endpoint-remapping bound from $O(\sum_i k_i g_i B_i)$ to $O(\sum_i (g_i + k_i \log(g_i+2))B_i)$.*

Proof. There are $2k$ domain or range intervals, so their union has at most $2k$ components and its complement has at most $2k + 1$ maximal gaps. Apply Theorem 7.1 to each contraction event independently. □

The corollary concerns endpoint transport, not every operation performed by the full weighted checker. Weight-run overlays, reflection and translation folds, source reconstruction, and geometric terminal checks remain separately charged. The existing polynomial-in-compressed-input AHT guarantee is not converted into a new bound for the overall unknot recognition problem by this local improvement.

7.3 Independent implementation and authentication order

The new private function `interval_orbit_verify._contract_pairings` constructs its own gap-end and prefix arrays. The unweighted checker calls it only after independently recomputing all static gaps and checking exact equality with the proposed list. The weighted histogram and transversal checkers call the same verifier-side helper only after the unweighted trace has already been authenticated. The helper imports no producer module and is not a public verifier entry point. Weighted transport continues to use its existing independent formulas; selector transport continues to restore original labels by reverse gap insertion. The inverse formula above explains why forward contraction and original-label restoration remain compatible.

The producer change is smaller. Its old union scan maintained both a rightmost endpoint and a continually rewritten list of merged occupied intervals. The new scan records an occupied interval only when its component ends, using one running start and one running right endpoint. Intervals that overlap or are adjacent remain one component; a strict separation emits exactly one maximal gap. Sorting order, gap order, endpoint mapping, returned pairing objects in the no-gap case, event statistics, and the certificate trace are unchanged. This reduces allocation and redundant integer operations without changing the producer's asymptotic sorting bound.

7.4 Source reuse inside one composed call

The original composed producer constructs source parallelity geometry in `normal_cut_parallelity`, then reconstructs the same geometry when asking for ordered boundary words. The revised producer passes its own already constructed geometry to a private boundary continuation. The public boundary-word API still performs full source preparation and coordinate validation when called on its own.

The checker has a separate implementation of this reuse. Its private parallelity replay returns the independently reconstructed source geometry, the independently recounted summary, and the source digest only after all four parallelity proofs have passed. The composed checker then passes that reference geometry to the private boundary replay, which verifies the two boundary traces and their source digest. The subsequent finite geometric checks consume the reconstructed reference geometry and summary. They no longer decode charts back out of the certificate after authenticating them. No cache survives a public call, and no public parameter accepts a prepared source, source geometry, or authentication token. There is consequently no cross-call cache-invalidation condition to maintain.

Proposition 7.3 (Extensional preservation under local reuse)**.** *For fixed source input and certificate, on executions that complete without cancellation or callback exceptions, replacing repeated deterministic source reconstruction by one invocation-local reconstruction leaves the acceptance predicate unchanged. Callbacks must not mutate the source or proof during the call. Reusing a producer's*

geometry within the producer and reusing independently reconstructed geometry within the checker do not introduce a producer-to-checker trust dependency.

Proof. Let $R(S)$ be the deterministic reference reconstruction of source S and let V_1, V_2 be the two existing replay predicates. The old predicate is $V_1(R(S), P_1) \wedge V_2(R(S), P_2)$; the revised predicate computes $G = R(S)$ once and evaluates $V_1(G, P_1) \wedge V_2(G, P_2)$. These are extensionally equal. All ordinary public calls still reconstruct from the actual source, and no reference object is supplied by either certificate or producer. The callbacks continue to propagate cancellation; a cancelled call returns no successful authentication and stores nothing for later use. □

This proposition compares completed executions. A callback that cancels on its jth invocation may cancel one version and allow the other to finish; equality of that behavior is not asserted. Both propagate cancellation and callback exceptions without authenticating the interrupted call.

The count of callback invocations is intentionally not an invariant: the removed duplicate work no longer needs callbacks, while prefix construction adds callbacks inside its own bounded loops. Certificate-operation budgets are invariant: the composed checker still totals all six supplied traces before replay, and the producer still shares its cycle allowance across all six discoveries.

7.5 Validation and limits

The focused validation includes existing interval-orbit, weighted-orbit, weighted-transversal, contraction, source-parallelity, boundary-word, handlebody and three-handle tests. New tests exhaust every retained/deleted mask for universe sizes 0 through 9, checking the forward map against a literal oracle and the inverse against the original retained labels. A further 700 deterministic random signed interval systems check both the producer union sweep and verifier endpoint map against naive arithmetic. A 2^{16384}-scale case confirms exact binary arithmetic without point expansion. Hostile tests cover wrong, reordered and overlapping gap lists, coordinate and source mutation between calls, matching false digest strings, mutated nested source charts, callback exceptions, exact operation-budget boundaries, and attempted injection of public prepared-source parameters. Producer helpers are deliberately disabled while independent checkers run.

The paired timing experiment uses the entire 106-case native source corpus, with separately imported pinned-baseline, source-reuse-only and final implementations. Every repeated result, statistic, census, and complete certificate is compared for exact equality. Method order is randomized within each case and repetition. The recorded timings are complete native cut-census production plus independent replay; they do not measure standalone unknot decisions and are not an asymptotic experiment. The report preserves all individual timings and source hashes, permitting inspection of small cases, orbit-dominated cases, and inconclusive calls separately.

8 Experiments, source audits, and what the measurements establish

8.1 Reproducible input and measurement protocol

The native input is the pinned 106-case marked-parallelity corpus. Its SHA-256 digest is recorded in `PROVENANCE.json` and in every native audit. The export reconstructs each source from the triangulation and normal coordinates, reconstructs it a second time through the independent reference implementation, and checks exact agreement before extracting any matrix. No precomputed geometric matrix is accepted merely because its filename says that it came from the source.

The timing experiment imports three separate namespaces: the exact pinned baseline, the source-reuse-only ablation, and the final speed patch. It runs five rounds for every case, randomizing the order of the three variants within each case and round with recorded seed 20261012. Each observation includes the native whole-cut producer and, on a complete result, the independent whole-cut certificate checker. The program compares the entire returned objects, including statistics, censuses, and certificates, on every repetition. All comparisons pass. There are 92 complete and 14 inconclusive outcomes in each timing variant.

The environment is CPython 3.12.14 on Linux x86-64, with `perf_counter_ns` as the timer. The raw observations, Python version, platform string, source hashes, and aggregation rule accompany the article. They are measurements on this host, not hardware-independent constants. Two subsequently corrected docstrings differ from the measured snapshot; their recorded hashes reproduce exactly and the executable abstract syntax trees are unchanged. The delivery provenance records both versions.

8.2 Full-query speed and a controlled ablation

Variant	Production (s)	Independent replay (s)	Full query (s)
Pinned baseline	32.2088	23.4109	55.7582
Source reuse only	32.1755	22.4004	54.8330
Final speed patch	27.3450	15.1712	42.6671
Baseline/final ratio	1.1779	1.5431	1.3068

Table 1: Sum, over all 106 cases, of the five-round median for each measured quantity. The median of a sum need not equal the sum of medians, so the production and replay columns need not add exactly to the full-query column. No lower-forest portfolio is included in these timings.

The aggregate full-query reduction is $1 - 42.667140636/55.758189326 = 23.48\%$. The median, over cases, of the within-case paired baseline/final ratio is 1.3004. These two summaries agree closely here, although they answer different questions: one weights large cases by their wall time, while the other describes a typical case. Source reuse alone changes aggregate full-query time by a factor of only 1.0169. The larger final gain is consistent with the profile's finding that orbit contraction and transport dominate this particular workload. It would be incorrect to attribute the entire improvement to caching source geometry.

A separate synthetic experiment isolates endpoint mapping and verifies exact equality with the preceding naive formula. It uses equal numbers of pairings and static gaps, from 64 through 4096, with five paired rounds. At 4096 pairings the median naive mapper time is 2.3172 seconds and the median prefix mapper time is 0.004870 seconds. Their ratio is approximately 476; the median of the five paired ratios is approximately 509. These are deliberately labelled mapper-only numbers. They are not the speedup of the native recognizer or even of its entire orbit checker.

8.3 Fresh native parameters and exact finite-core outcomes

The native exporter checks the raw unsigned occurrence bound on every one-handle column: total at most three, and at most two for a corner handle. It also exports the canonical lower forest and a forest attaining $\kappa = |E_3| - r_G(E_3)$. The resulting distributions are:

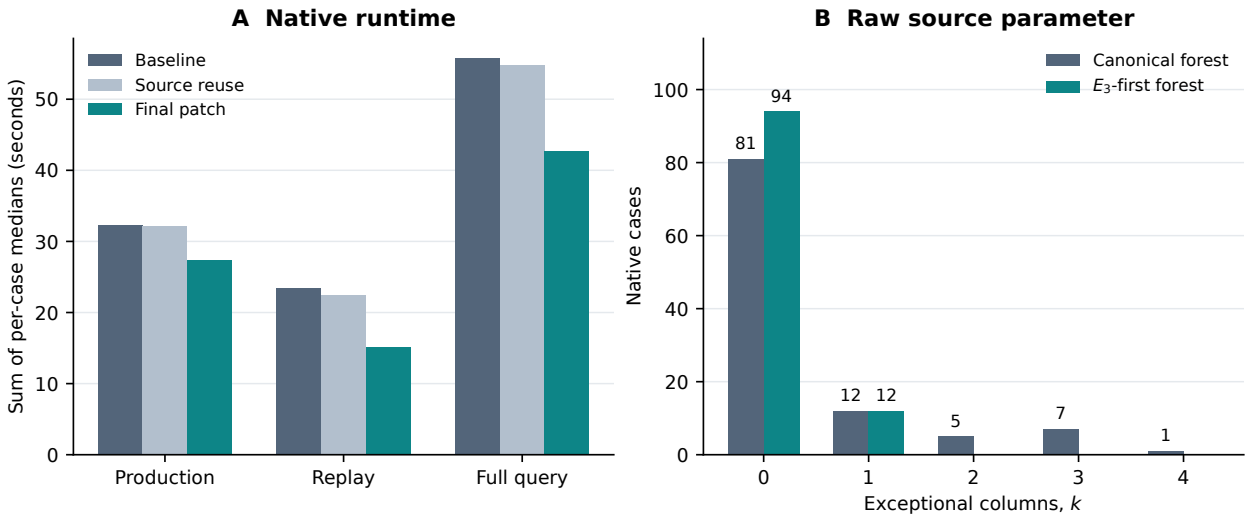


Figure 1: Two distinct native results. Left: complete-query timing of the behavior-preserving speed patch. Right: distribution of the raw exceptional-column count before and after the proved lower-forest parameter optimization. The right panel is a source audit, not a timing experiment.

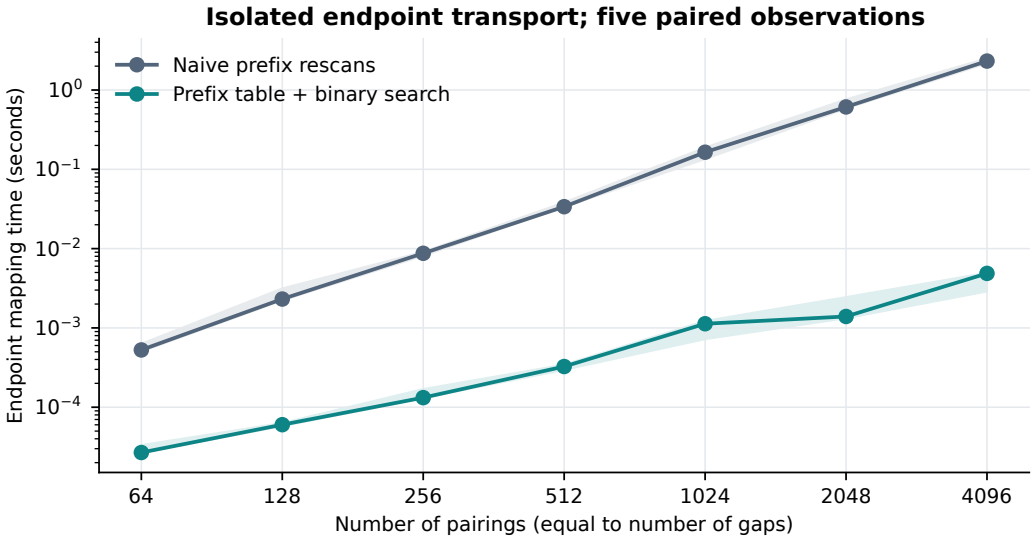


Figure 2: Synthetic endpoint-remapping measurements, with medians and the range of the five observations. Both axes are logarithmic. The theorem establishes the asymptotic improvement; this experiment isolates the implementation that realizes it. Source authentication and other orbit operations are outside this plot.

Lower-forest policy	$k = 0$	$k = 1$	$k = 2$	$k = 3$	$k = 4$
Canonical	81	12	5	7	1
E_3 first	94	12	0	0	0

For the canonical matrices, 89 cases have no nonfree special SCC, seven have one, eight have two, one has three, and one has four. Their ideal counts are respectively 1, 2, 4, 8, and 16. In this raw corpus the surviving special-SCC orders happen to be antichains, so the ideal refinement does not further reduce the Boolean state count. It is a proved structural improvement whose benefit requires a different order structure. After E_3-first optimization, the nonfree dimensions are zero in 94 cases, one in ten, and two in two; the largest ideal count is four.

Finite supplied-matrix result	Canonical lower forest	E_3 first
Old two-row solver: success	81	94
Old two-row solver: unsupported/inconclusive	25	12
Exact history solver: success	100	106
Exact history solver: finite infeasibility	6	0
Deterministic width solver: success	100	106
Randomized ideal solver: success	100	106

Table 2: The native *raw core* finite problems. The six unsuccessful canonical cases yield exact finite negatives in deterministic search and inconclusive outputs in randomized discovery. Every positive certificate passes the separate finite verifier. These counts are not whole-cut handlebody counts.

All 212 finite instances—106 with each lower-forest policy—were also checked by an independent exhaustive spanning-tree and literal-peeling oracle. It enumerated 136 spanning trees for each policy and agreed with the exact solvers on every instance. The native dual rank is at most one in this corpus. Thus this experiment authenticates a useful native domain and detects real lower-forest effects, but does not empirically stress the new theorem at large dual rank. In particular, all 106 optimized positives are already found by the deterministic mandatory-group candidate followed by peeling; none needs a nonzero determinant discovery. The algebraic implementation is exercised separately against explicit basis expansions.

The six raw-core rescues are the pilot Fibonacci cases with indices 04, 06, 08 and 10, together with `pilot/lst_4_9/0` and `pilot/finite_trefoil/0`. All 100 prior raw-core successes are preserved by the optimized forest on this corpus. That empirical preservation is not asserted as a theorem for arbitrary inputs.

8.4 Five additional whole-cut certificates

A separate, optional producer portfolio first tries the maintained planner and then two bounded lower-forest policies: E_3 first and reverse edge order. The finite core proof is replayed, and any completed whole-cut packet is submitted to the unchanged public whole-cut verifier. The adapter uses a private producer-module instance; it does not replace a public module's planner or alter the certificate schema.

New full-certificate case	Successful alternative
`pilot/fibonacci_lst_04/0`	E_3 first
`pilot/fibonacci_lst_06/0`	E_3 first
`pilot/lst_4_9/0`	E_3 first
`pilot/finite_trefoil/0`	E_3 first
`pachner23_5_8_2/mixed`	Reverse order

Each new packet contains the original source, coordinates, and complete native certificate, with its hash recorded in the audit. Combining these five verified positives with the maintained 92 gives 97 complete whole-cut censuses out of 106. The original speed patch continues to produce exactly the original 92 outputs; the coverage increase belongs to the separately dispatched research portfolio.

The larger Fibonacci cases with indices eight and ten now pass raw core cancellation but stop at annular attachment. The finite figure-eight positive target still stalls in the whole-cut core path in the bounded search. These observations identify a geometric continuation problem, not a reason to relax unsigned cancellation or certificate checks.

A historical Regina oracle audit, tied to this same corpus hash, identifies eight genuine all-handlebody misses among the original fourteen refusals; the other six contain at least one non-handlebody piece. This classification is explicitly labelled historical in the package; it was not recomputed as part of the timing run. The three saved interface obstructions in the preceding forest report belong to the latter category. Seeking geometric moves that preserve the whole cut yet turn all of their components into handlebodies would therefore be a misguided positive target. If sphere capping is intended, it needs its own changed terminal specification and certificate.

8.5 Independent finite identities and hostile inputs

The new unit suite has 15 test methods. Its exhaustive family covers all 4096 combinations of a binary 3×2 matrix and three edge records chosen from two loops, one eligible edge, and an unavailable edge. It also checks 320 weighted instances with wider exceptional columns, 180 small posets including chains and antichains, all 256 three-variable Boolean truth tables, resource limits, and certificate/source mutations.

The independent mathematical audit programs add the following checks:

- 800 matrices, 2836 tree/matrix pairs, and 46 127 exceptional schedules checked against literal peeling;
- 700 covering instances and 1377 determinant evaluations compared with direct common-basis polynomial expansion;
- 650 Boolean-sieve matrices, 2300 tree/matrix pairs, 8657 activation table entries, and 711 determinant-versus-basis comparisons;
- 1800 random posets on one through nine elements, containing 47 467 enumerated ideals, with direct zeta, inverse, maximal-element and filter identities checked;
- 650 further weighted instances, with up to five rows in an exceptional support, checked directly against the delivered width and randomized implementations and an independent tree/peeling oracle.

The focused maintained-runtime regression passes 100 tests, and the separate portfolio suite passes nine, for 109 distinct implementation tests in total. These are additional to the 15 finite-solver

test methods. The runtime suite includes the independent gap arithmetic and source-mutation checks described in section 7. Test identities, seeds, commands and raw reports are included so that a reader can distinguish proof, finite testing, and a native source-authenticated result. Finite audits do not replace the proofs, and passing selected runtime tests does not claim that every test elsewhere in the large repository was rerun.

9 From local algorithms to a global complexity argument

9.1 An explicit ledger of the remaining obligations

A local algorithm can be fixed-parameter tractable while the enclosing recognizer remains exponential. There are four independent quantities to control: the cost of producing a legal next surface or move, the number of states visited, the encoded size of each state, and the cost of resolving the terminal topology. A small matrix at one successful cut gives no bound on the other three quantities.

The reviewed hierarchy preprint supplies a polynomial compressed-cutting primitive for handle structures of uniform type, polynomial in handle count and logarithmic extended weight. It returns the complement of the parallelity bundle and the bundle topology and attachment data (Theorem 14.4), while Section 9 separates a step-count analysis from a full runtime estimate [15]. This is the appropriate interface for the present work: geometry must arrive in a compressed, source-authenticated representation, and its attachment information must survive every simplification.

Obligation	Result available here	Still required globally
Raw local completion	Exact J-state reduction; deterministic width algorithm	Completeness of the chosen geometric move class at every needed state.
Representation size	No expansion in multiplicity in these finite operators	Bounds on handles, charts, runs, and integer bit lengths after all cuts and transports.
Hierarchy control	Explicit local cost parameters	A bound on total states, restarts, and surface-production work.
Boundary and bundle data	Existing native verifier; five new accepted packets	Complete primitive-attachment, puncture, and terminal handling where current certificates refuse.
Randomness	One-sided local witness misses, exact positive replay	A global error schedule, or a deterministic structural regime.

9.2 A conditional composition bound

The following accounting statement makes the target testable. It is a conditional composition result, not a theorem that the hypotheses hold for all knot diagrams.

Proposition 9.1 (A sufficient deterministic route to quasi-polynomial time)**.** *Suppose a correct and complete recognition algorithm on n-crossing input diagrams has the following independently established properties:*

1. *it visits at most $Q(n)$ states and makes at most polynomially many local completion calls per state;*

2. *each call is equivalent to the finite completion problem of this article, with explicit encoded size* $N \le n^b$ *for a fixed* b*;*

3. *all remaining production, transport, and verification work per state is bounded by* n^b *after increasing* b *if necessary;*

4. *the implication width of every local call is at most* $w(n)$*.*

If $Q(n) \le \exp(O((\log n)^a))$ *and* $w(n) = O((\log n)^c)$ *for fixed* $a, c \ge 0$*, then replacing its finite completion calls by the deterministic width algorithm gives total time*

$$\exp\Big(O\left((\log n)^{\max\{a,c+1,1\}}\right)\Big).$$

In particular it is quasi-polynomial. The correctness and completeness assumption includes all geometric and terminal decisions, not only the existence of the local certificates considered here.

Proof. For $w > 0$, the ideal bound gives $J \le (d+1)^w \le (N+1)^w$. The deterministic width algorithm therefore costs $N^{O(w+1)}$ per call. For $w = 0$ it is polynomial as well. Substituting $N \le n^b$ gives $\exp(O((w(n)+1)\log n))$. Multiplying by the permitted number of states and calls and adding the other work yields the displayed bound. Because each local algorithm solves exactly the specified finite problem, this replacement does not alter an independently proved global correctness argument. □

The randomized alternative replaces the local factor by $2^d \operatorname{poly}(N)$, or more precisely $J \operatorname{poly}(N)$. With a known upper bound on the number of relevant calls, distribute a desired total miss allowance ε among them and repeat independent identity tests accordingly. A union bound controls the probability of any required local witness being missed, provided all other completeness obligations are met. A zero test still cannot be treated as a deterministic negative certificate.

For example, polynomial state size and $d = O(\log n)$ give a polynomial randomized local cost. Polynomial state size and $w = O(\log n)$ give a deterministic local cost $\exp(O((\log n)^2))$. If a hierarchy visits $\exp(O((\log n)^3))$ states, the total bound remains quasi-polynomial but has the larger exponent. This explains why an article must specify whether it controls local work, the state count, or both.

The bound $k \le 2T$ for a raw core with T tetrahedra is insufficient on its own: 8^{2T} is exponential in T. Nor does observing $\kappa \le 1$ on 106 inputs establish such a bound on an infinite class. The useful advance is that there are now precise quantities—κ, d, J, w, and q—for a global geometric theorem to control.

10 Proposed research program

The next projects should have a definite theorem, failure example, or replayable artifact as their output. The order below emphasizes the two current positive gaps in annular attachment, then the structural questions most directly connected to asymptotic complexity.

10.1 Geometric continuation and choice of lower forest

1. Complete the two newly exposed Fibonacci attachment cases. The index-eight and index-ten Fibonacci cuts now admit raw core cancellation under the optimized lower forest, yet their annular interface stage refuses. Export the exact remaining annular curves, the lower-forest basis, and all source endpoint maps. Determine whether a verified change of meridian system makes

the attachment primitive in the required geometric sense. A useful result is a family theorem with a compressed construction, not just two extra successes. The deliverable should be a source-bound move sequence followed by an unchanged terminal check, together with a bound in layer count and coordinate bit length.

2. Optimize lower forests for completion, not only for k. The graphic-nullity formula solves the parameter objective exactly, but minimizing that objective need not preserve a successful attaching pattern. Study the joint choice of lower and top forests. First search for an explicit native example where every minimum-k lower forest fails but another lower forest succeeds. If such examples exist, identify a secondary objective or a bounded family of exchanges that preserves completeness. If they do not occur in a well-defined native subclass, prove the exchange property. The reverse-order rescue of the Pachner case already shows why a single numerical score should not silently replace a portfolio.

3. Authenticate geometric tightening and handle slides. Actual unsigned occurrences can exceed the algebraic obstruction after curves are tightened or the meridian system changes. Signed exponent sums, gcd tests, or free-group simplification alone do not provide the geometric disc required by the current certificate. Develop a compiler for an embedded boundary-curve isotopy or a geometrically realized handle slide, with certified endpoint transport and intersection counts. Existing polynomial compressed-curve algorithms supply relevant primitives [16]; the open work is their integration with this marked source representation and simultaneous annular attachments. The first milestone is a checker for one move, including its inverse and the preservation of boundary pattern and orientability.

4. Specify punctured-handlebody terminals separately. Six original refusals contain non-handlebody pieces according to the historical oracle. Some applications may instead want a handlebody after certified sphere capping. Define that terminal class explicitly, including which boundary spheres are capped and how the original component is recovered. A new puncture certificate should not be accepted as an ordinary handlebody certificate. This project can turn otherwise misleading "misses" into a useful terminal specification while preserving the logical meaning of the existing genus census.

10.2 Sharper structural parameters and exact algorithms

5. Exploit native cycle compatibility of triple supports. The current theorem permits arbitrary exceptional supports. Source boundary compatibility imposes additional mod-two cycle conditions in the dual graph, as audited in the preceding normalization report [20]. A three-edge even subgraph has very restricted forms, involving triangles, parallel edges, and loops. Determine whether these constraints give a stronger exchange theorem, a smaller effective parameter, or a polynomial class beyond support two. The target is an exact reduction respecting actual unsigned entries; parity can constrain the search but cannot substitute for a unit crossing. Any hardness construction used to delimit this problem must satisfy the native source constraints before being called a topological hardness result.

6. Saturate unary consequences before building the poset. Forward chaining from one row may imply another through several nonlinear rules. Such a unary logical consequence is safe to add as an ordinary implication if its derivation is retained. Investigate whether repeatedly adding these

consequences substantially collapses the special-SCC poset, and prove the exact correspondence with the original closure semantics. The obvious polynomial preprocessing tries closure from each row. The interesting questions are whether a smaller set of representatives suffices, how much J or w can decrease, and whether native zero patterns impose a uniform bound on the saturated width. The positive checker can continue to replay the final tree and pivots in the original matrix.

7. Determine the best exact route when q, w, and d differ. The present alternatives have costs $J\,\mathrm{poly}(N)$ with one-sided randomness, $J(m+1)^w\,\mathrm{poly}(N)$ deterministically, and $2^q J\,\mathrm{poly}(N)$ by exact counting. Implement the Kirchhoff alternative with fraction-free or certified modular determinants, then compare these regimes using source-derived parameters. Seek a deterministic $2^{O(d)}\,\mathrm{poly}(N)$ algorithm when q is large. The signed common-basis polynomial is the concrete difficulty; a positivity argument for a different polynomial must prove that it has exactly the required support. General nonnegative-polynomial detection results are relevant context, but do not directly apply to the mixed-sign common-basis enumerator [17].

8. Replace large explicit ideal tables when the frontier is small. A poset can have many ideals even when a particular elimination ordering has a small active frontier. Study a decision diagram or a dynamic program that compiles the activation function and ideal constraints together. Specify exactly what boundary information must be retained so that a tree outside the processed region is represented faithfully. It is not enough to compress the Boolean function if the graphic common-basis constraints lose their dependence across the boundary. A useful theorem would express time in an explicitly defined width of the combined dependency and graphic structure, with an independent witness extractor.

10.3 Verification, representation size, and the hierarchy

9. Produce portable negative finite certificates. The deterministic search can exhaust its finite space, but most of its negative outputs currently have no exported proof object. Develop a certificate that an independent checker can validate without trusting the search program. Possible starting points are per-state graphic rank obstructions, a shared acyclic proof of all failed representative choices, or exact Kirchhoff filtered counts in the small-q regime. The certificate must remain bound to the original graph and matrix, and its verification cost and size should be included in the parameter bound. Randomized zero evaluations are insufficient.

10. Improve the remaining compressed transport costs. The prefix theorem removes repeated gap sums, but the measured runtime still includes weight-run overlays, translations, reflection folds, selector restoration, and repeated trace traversals. Profile these after the current patch. Investigate sorted batched endpoint transport and interval-run composition, with explicit bit bounds and retained/deleted endpoint conventions. A proposed improvement should preserve independently authenticated gap lists and demonstrate a gain on complete native replay. The corpus's 2^{16384} coordinate scale is a useful check that a convenient array representation has not introduced point expansion.

11. Build native stress families for the actual new parameters. The present raw corpus has dual rank at most one and very small ideal counts. Create triangulated source families with increasing dual rank, controlled exceptional-column cycles, nontrivial implication chains, and large binary coordinates. Report T, row count, dual rank, k, κ, d, J, w, q, certificate size, and replay

time. Keep abstract graph/matrix generators separately labelled: they are excellent logical tests but do not establish realizability by a triangulated source. Prioritize families that distinguish an exponential number of activation masks from a polynomial number of ideals.

12. Prove a hierarchy-wide complexity budget. Track the number of active nonparallel handles, boundary-pattern records, interval runs, restart events, and coordinate bit lengths throughout a complete hierarchy implementation. Seek an amortized invariant bounding both the number of visited states and the maximum local ideal complexity. A decrease in handle count alone is insufficient if a restart duplicates many states or an attachment representation grows too quickly. The desired deliverable is a theorem supplying the hypotheses of proposition 9.1, or a counterexample showing which proposed potential fails. This is the central remaining bridge from the proved local operators to Lackenby's announced asymptotic target.

10.4 Recommended next integration checkpoint

The behavior-preserving speed patch is ready for ordinary code review and the repository's usual integration gates. The lower-forest portfolio is separate and can be reviewed with its five full packets and shared resource limits. The exceptional-row solvers should enter as research operators with measured dispatch parameters, explicit state/work limits, and the unchanged finite verifier. Their source adapter must carry all lower-forest and attachment provenance before they are used as native geometric claims.

For the next research iteration, the highest-value concrete target is the source-authenticated annular continuation of the two larger Fibonacci examples. In parallel, the structural priority is to test and then prove a bound on implication width after geometric simplification. Those tasks would connect the native coverage improvement and the new deterministic complexity theorem at an interface where both can be checked precisely.

A Reproduction, integration, and the certificate boundary

A.1 Package structure and source pin

The package is intended for review before incorporation into ProveIt. The LaTeX article, source fragments, vector figures, executable algorithms, independent audits, raw measurements, source-bound packets and patches are all included. The root `README.md` gives exact commands and the recommended patch order. `PROVENANCE.json` identifies the repository pin, the five inspected incoming archives, and their hashes; `MANIFEST.sha256` inventories the delivered files.

The main LaTeX source is `article/exceptional_rows_and_certificate_replay.tex`. Compile from its directory with:

```
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  exceptional_rows_and_certificate_replay.tex
```

The source uses standard LaTeX packages. Its figures are included as vector PDFs; rebuilding them uses Python, NumPy, and Matplotlib. The finite forest solvers and their independent checker use only Python's standard library. Native reproductions use the dependencies of the maintained ProveIt fast package.

A.2 Finite problem schema and available algorithms

The finite input is a JSON object with exactly four fields:

```
{
  "vertices": 2,
  "columns": 2,
  "edges": [[0, 1], [0, 1], [0, 0]],
  "rows": [[[0, 1], [1, 1]], [[0, 1], [1, 2]], [[0, 1]]]
}
```

Row i has edge record i. A missing eligible edge is encoded by `null`; loops are also ineligible. Each sparse row lists distinct column indices and strictly positive integer counts. Booleans are not accepted as integers. Root vertex zero is distinguished. Empty matrices, isolated source vertices, large integer labels and multiplicities, repeated edges and invalid records have explicit validation behavior.

Entry point	Search	Negative convention
`solver.solve`	Exact exceptional schedules, support at most three for useful columns	Exhaustive `NO_FOREST`; limits are inconclusive.
`solver.solve_randomized`	Reference schedule plus covering sieve	A randomized zero is inconclusive.
`nonlinear.solve_boolean`	SCC/ideal determinant sieve, arbitrary exceptional support	A randomized zero is inconclusive.
`nonlinear.solve_width`	Exact streamed ideals and minimal-frontier representatives	Exhaustive `NO_FOREST`; limits are inconclusive.
`base_verify.verify`	Independent tree and original-matrix replay	Accepts only supported proof objects.

Every success carries a tree and complete peel sequence. The public discovery functions construct these by exact integer operations; callers that rely on the result must invoke the supplied independent checker. The tests and audit programs do so for every reported positive. They do not regard a discovery program's own assertions as independent verification.

The implementation stores ideal coefficients as integers. The formula

$$c(A) = \sum_{S \subseteq \max(A)} (-1)^{|S|} g(A \setminus S)$$

bounds their absolute values by 2^d, so the coefficient storage is $J \operatorname{poly}(N)$ bits. The sharper $O(2^d \log N)$ theoretical table bound in the Boolean theorem uses residues modulo the prime; the delivered integer tables still satisfy the stated $J \operatorname{poly}(N)$ ideal bound.

The fixed default prime is $2^{61} - 1$. The implementation checks primality within an explicit 64-bit domain and requires the chosen prime to exceed twice the number of eligible edges when constructing the determinant sieve. Thus the prime characteristic exceeds every copy count and provides distinct Vandermonde labels. For arbitrary asymptotic input sizes, the mathematical algorithm chooses a prime in a polynomial interval rather than relying on this fixed implementation constant.

An omitted random seed uses uniform system randomness. A supplied seed makes experiments reproducible; a probability bound is justified by the uniform independent sampling model, not by one particular pseudorandom seed. Once an evaluation is nonzero, extraction uses no further randomness. Resource caps return `INCONCLUSIVE`. Most exhaustive deterministic negative answers do not yet carry a portable negative proof, as their `certificate` field and evidence string make explicit.

A.3 A minimal example that needs exceptional propagation

The JSON example above has rows a, b, c and unsigned matrix

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 0 \end{pmatrix}.$$

The first column is exceptional. The second gives the ordinary implication $b \to a$. Rows a and b are parallel eligible edges; c is a root loop. Deleting a as the spanning tree leaves two owners of the triple column and a coefficient two in the other column, so peeling stalls. Deleting b instead allows a to peel on column one and then c on column zero. The certificate is the tree [1] and peel sequence $[(0, 1), (2, 0)]$.

The special-row poset has relation $b \prec a$ and isolated element c. It has six ideals rather than eight Boolean subsets. This example makes the conjunction and order-ideal reduction visible in a small input; the general results are justified by the proofs and the separately authenticated native experiments, not by the example alone.

A.4 Counterexamples that constrain the algebraic implementation

Future heads cannot satisfy their own premises. Take a one-vertex dual graph, so no row can be deleted by a tree, and the matrix

$$\begin{pmatrix} 1 & 2 & 2 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}.$$

The first column is a triple; the ordinary implications are $a \to b$ and $a \to c$. There is no initial seed and no legal first peel. A proposed exceptional head a cannot be used in advance to justify its prerequisites b, c. This rejects the tempting but invalid replacement of the prefix head set H_{i-1} by the final head set in the schedule reduction.

Different copies need different variables. Consider a rank-two star with two edges e_1, e_2, both in one mandatory group. Its copied common-basis polynomial has the form

$$z_{1,\mathrm{rep}} z_{2,\mathrm{fill}} - z_{1,\mathrm{fill}} z_{2,\mathrm{rep}}.$$

There is a valid spanning tree. Identifying representative and filler variables for each original edge makes the polynomial identically zero. The independent variable for each copy is therefore part of the theorem, not an implementation convenience.

One deletion scan is insufficient over a field. For three parallel edges in rank one, take $Q = x_1 + x_2 + x_3$ over $\mathbb{F}_7$ and evaluate at $(1, 1, 6)$. The total is one. Removing the first or second variable yields zero; removing the third leaves two. A single scan can therefore stop with two variables. A second scan extracts one. The repeated-scan implementation tests this exact regression.

Field cardinality does not replace characteristic. Over any characteristic-two extension, the same polynomial evaluated at $(1, 1, 1)$ is nonzero, while every one-variable deletion is zero. Euler's identity then only gives $3 \equiv 1 \pmod 2$. Arbitrarily large extension fields do not fix this example. The extraction lemma needs characteristic greater than the initial number of variables.

Inverse ideal transforms need reverse topological order. For a chain $a \prec b$, the coefficient of its full ideal is $g(\{a, b\}) - g(\{a\})$. Processing a before b in the subtraction transform introduces an erroneous $g(\varnothing)$ term. A regression also labels ordinary rows in the reverse of their reachability order, ensuring that sorting row identifiers cannot masquerade as topological sorting.

A.5 Evidence levels and implementation status

Claim	Evidence delivered	Boundary
Finite schedule/ideal/width theorems	Mathematical proofs; independent finite audits; executable solvers	Supplied rooted graph and unsigned matrix; not a general handlebody theorem.
Kirchhoff exact counting	Complete proof with bit bound	Not a separate delivered optimized solver.
Native raw arity and optimal k	Local source proof; fresh source/reference equality checks	Before arbitrary bundle augmentation or slides.
Prefix contraction	Exact quotient/inverse proof; independent finite tests	Endpoint substep, not the whole orbit algorithm.
23.48% full-query reduction	Five paired rounds; complete returned-object equality on 106 cases	Recorded host and workload only.
Five new whole-cut positives	Original-source packets; unchanged public verifier acceptance	Bounded two-policy portfolio, not exhaustive forest search.
Historical oracle split	Prior report and identical corpus hash	Supporting classification, not a fresh oracle run or the new proof.
Global quasi-polynomial recognition	Conditional accounting proposition and explicit research obligations	Not established in this continuation.

The mathematical proofs are conventional written proofs, accompanied by independent executable checks. They have not been formalized in Lean. Natural formalization targets are the closure equivalence, source-SCC criterion, ideal inversion identity, and fixed-evaluation extraction lemma; each has a small explicit interface independent of three-manifold geometry.

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